L07 – Physics-informed Neural Networks (PINN)

Foundations Lecture Series

Course Instructor

Microgrid 3-phase Security AI Project

Foundations Lecture 7

Keywords: physics-informed loss, soft constraints, limit-aware learning, Raissi 2019

Lecture roadmap

1. The PINN idea – physics as a soft constraint
2. Loss design: data + physics
3. Strict vs soft constraints
4. Limit-aware learning – the project's flavor
5. When PINNs help and when they don't
6. Connection to the project's Week 6 PI-MLP

Prerequisite

L05 (MLPs, loss design, PyTorch). Helpful: L01–L04 for the physics being constrained.

Where PINNs come from

TODO: *Briefly trace the lineage from prior physics-based ML to Raissi-Perdikaris-Karniadakis 2019. State the central idea: encode the known PDE/ODE/algebraic constraint into the loss function so the model is penalized for violating physics, not just for fitting data poorly.*

Why bother with physics in the loss

TODO: *Three reasons: (1) extrapolation beyond the training distribution, (2) data efficiency, (3) interpretability. Concrete example from microgrid security: a model that respects voltage limits "for free" is more trustworthy than one that learns them from data alone.*

Data loss + physics loss

TODO: *Write $\mathcal{L} = \mathcal{L}_{data} + \lambda \mathcal{L}_{phys}$. Explain that $\mathcal{L}_{phys}$ is typically the squared residual of a physical equation evaluated at the model's prediction. Discuss the role and sensitivity of λ.*

Multi-task framing

TODO: *Many PINNs in power systems are really multi-task: classification + regression + ranking + limit penalty. Show the project's Week 6 PI-MLP composite loss as a concrete example.*

Choosing physics terms

TODO: *Concrete physics terms relevant to this project: voltage limit penalty, current limit penalty, VUF penalty, power-balance residual, sequence-component consistency. Note that not all of these are strict PINN-style PDE residuals – some are just engineering limits.*

Soft constraints (the usual PINN)

TODO: *Add a penalty to the loss. Model can still violate the constraint; the penalty just makes that costly. Easy to implement, gradient-friendly. The default approach.*

Strict constraints (projection / hard layer)

TODO: *Project the output onto the feasible set, or build a network architecture that cannot violate the constraint by construction. Harder to design, often necessary for safety-critical applications.*

Which one for screening?

TODO: *For pre-screening (the project's framing), soft constraints are usually fine – we still re-run the simulator on flagged contingencies. For deployment without simulator backup, you would want hard constraints or conformal prediction.*

The project's flavor of physics-informed

TODO: *The project does not encode the full AC power-flow residual into the loss. It uses "limit-aware" soft penalties: voltage limits, current limits, VUF threshold. This is a pragmatic subset of full PINN. Be honest with students about this.*

Severity regression as physics-informed

TODO: *Predicting not just label (safe / unsafe) but also severity is a form of physics-informed multi-task learning. The severity head learns continuous physical quantities, which can stabilize the classification head.*

Ranking loss as physics-informed

TODO: *The project also adds a pairwise ranking loss: for top-k screening, the order of risk scores matters more than absolute probabilities. Frame this as another physics-aware multi-task component.*

When PINNs help

TODO: *Sparse data, known physics, extrapolation, safety-critical applications. Mention published cases from fluid mechanics, biomechanics, and power systems.*

When PINNs do not help (or hurt)

TODO: *Lots of data + well-tuned baseline. Wrong physics (e.g., a constraint that conflicts with the actual data-generating process). Overweighted penalty term. Poor optimization landscape from competing loss terms.*

What you will see in Week 6

TODO: *Show the high-level PI-MLP loss decomposition: BCE + severity MSE + component-margin + consistency + ranking. Promise to walk through each term. Forward-reference the ablation table from the paper.*

Homework

TODO: *Suggest: take the MLP from L05 (e.g., trained on a small N-1 dataset), add a voltage-limit penalty term, retune λ, and report whether recall and precision improve. Plot loss components over training.*

Exit checklist

TODO: *Items: (1) write the generic PINN loss form, (2) distinguish soft and strict constraints, (3) list the 5 physics terms used in the project, (4) name one scenario where PINNs help and one where they don't.*

References

TODO: *Raissi, Perdikaris, Karniadakis 2019 (JCP); Karniadakis et al. 2021 Nature Reviews Physics survey; Krishnapriyan et al. 2021 (NeurIPS, on PINN failure modes); Recent power-system PINN surveys (Misyris et al., Stiasny et al.).*